

## DATA ANALYSIS, ITERATION, AND FUNCTION

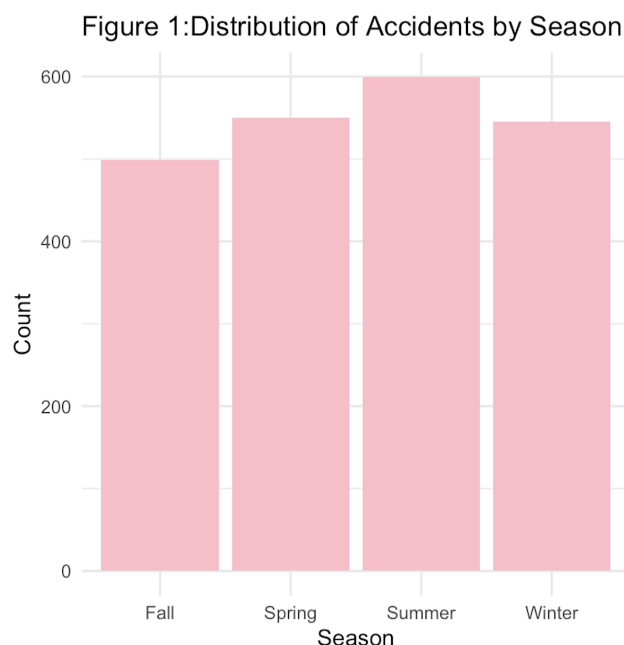
### Introduction

In this analysis I will conduct an exploratory data analysis or (EDA). I will investigate a group of variables relative to railway incidents and accidents. These variables include time of the day, season, weather condition, or engineer impairment, and other possible outcomes. And as part of the assignment prompt I will be considering their narrative fields (nar1-nar15) that have useful information, not available elsewhere. To make this report and analysis possible along the way I cleaned, grouped, and reshaped as needed.

### Data

I used the dataset of railway accidents and incidents provided in the assignment prompt. I also worked alongside the data dictionary to understand and use the meaning of key variables in my code and logic throughout the process. To begin I cleaned and reorganized the data, in categories I could use such as season, weather condition, and engineer impairment. I combined accident records into summary categories for statistics and visualizations. Following the workflow from a previous class lecture, like working with least terns dataset, I used this dataset looking for distributions, missing values, correlations or unusual patterns. The intention of this analysis is to explore, practice, and be described.

### Season

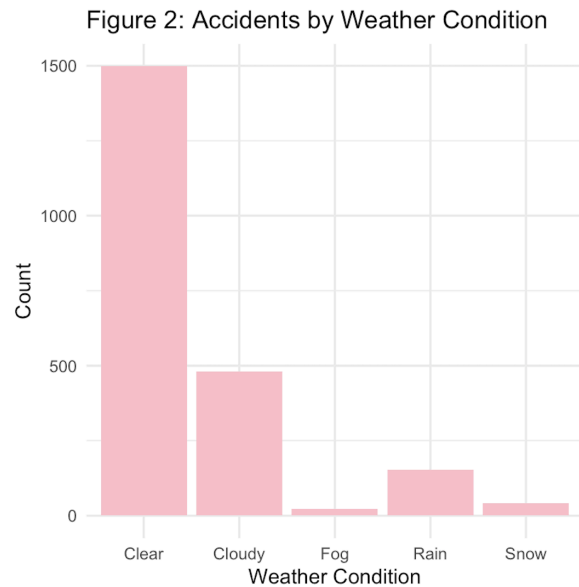


**Figure 1. shows the distribution of accident incidents by season.**

The figure displays the seasonal distribution suggesting that accidents/ incidents occur almost evenly in all four seasons. Although the summer appears to have the highest count of accidents, and the lowest numbers in the fall. Spring and Winter are in between these two extremes This means that there is a moderate seasonal variation in accidents/ incidents. Although the differences are not dramatic, we can assume summer is the one season that explains the most accidents. Seasonal patterns can be caused by changes in circumstances and activity in traveling, maintenance, environmental conditions, and heat and cold effects. But it's

important to conclude that these raw counts, and they do not immediately mean that this one season (summer) is more dangerous.

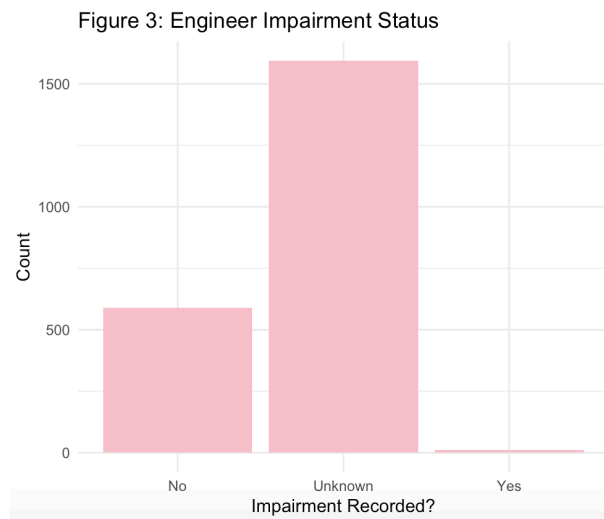
## Weather Condition



**Figure 2. shows distribution of accidnet/incdidnet by weather condition.**

Weather conditions show as one of the most common patterns in this data set. Most accidents/ incidents happened during clear weather. Cloudy conditions are the next common category with more accidents. Rain, snow and fog appear much less often, we can assume people are more aware and careful of their driving. The figure is helpful to understand the distribution of accidents across the different seasons.

## Engineer Impairment

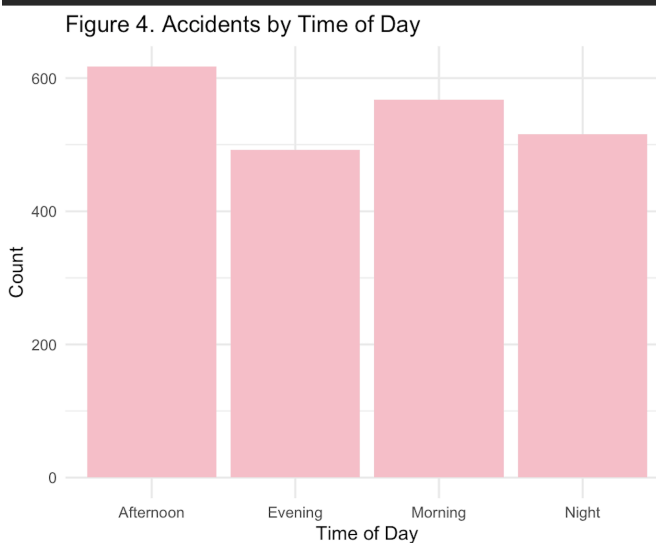


**Figure 3. shows recorded accidents/incidents caused by engineer impairment.**

This is another required variable to analyze, engineer impairment which is very limited because it is missing a lot of data. The summary also shows a large number of unknown/ missing values for alcohol and drug impairment. There are only a few “yes” cases. This also means that the dataset does not have enough data to make a credible and strong claim about how impairment is relative to accident. But this is an important finding because impairment is not always recorded

and this column is very incomplete. So instead of assuming that impairment is not relevant it's safer to make the conclusion that the dataset is not reliable for this variable/question.

## Time of Day

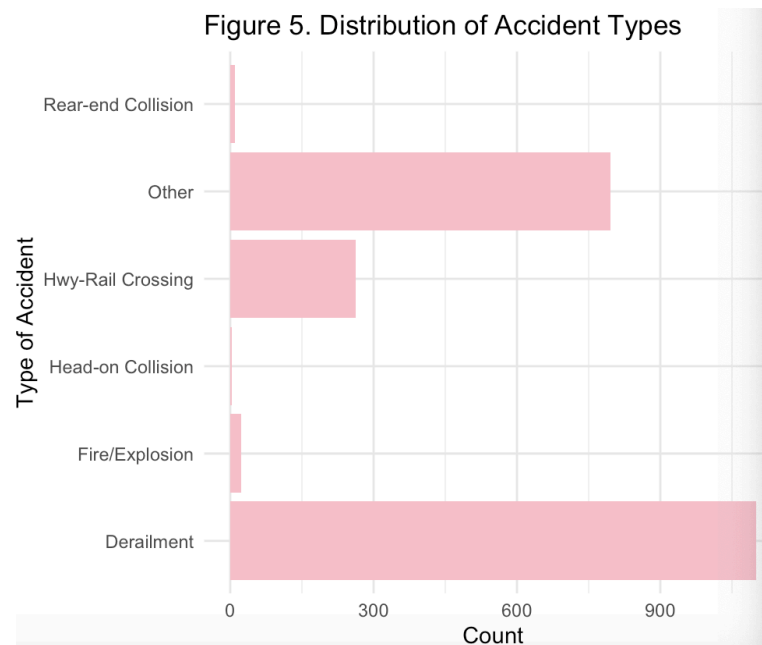


**Figure 4. shows the distribution of accident/iniidnet through the time of the day.**

Analyzing time of day was able to give me an idea whether accidents were spread through the day or during specific hours. The figure above shows how afternoon has the highest count, morning is second highest, night follows and evening is last. There are small differences, not extremes between these different times of day. The dataset is specific about including all parts of the day being recorded. During an accident, there is no even distribution.

Although this shows afternoon and morning have the highest activity for accidents, it does not prove that this time of the day makes accidents occur, or is more dangerous. It can be a correlation of things like energy levels, hours of traffic, and visibility,

## Potenetail features



Other potential features that may contribute to accidents include accident type and severity. Accident types can show what type of railway accidents are more common, and highlight some categories in this dataset. Such as derailment and other as shown. It can also help identify the patterns in season, weather and time of day, it is also very easy to use for this type of analysis. For the severity variable, this helps think about how many accidents happened. This can show if conditions are related to more serious outcomes. And in this case, derailment has a high count.

## **Narratives**

The narrative fields of nar1-nar15 have useful qualitative details not included in structure variables. These fields have more information about mechanical failures, environment, procedures, and unusual circumstances. Narratives also help explain cases where structure variables are missing, vague or not specific enough. Simple reviews of narrative text show repeated themes that appear that are not recorded elsewhere. It also means those narrative columns have good sources for future analysis.

## **Conclusion**

In conclusion, the EDA confirms that accident/ incidence patterns are more by season and weather conditions. Time of day is also an important variable because of activity data. Impairment variable and data is not substantial due to many missing values, which prevents making analysis and conclusion. One of the most important findings of this dataset is the many limitations it has. For example in a future analysis it could improve by comparing accident with exposure, going deeper into time of data and examining severity more.